

Sean Tran

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Education

The University of Texas at Dallas	B.S. in Physics (Expected Dec 2026) — GPA: 3.3	2024 – Present
	B.S. in Computer Science	2019 – 2024

Relevant Experience

NSF REU Undergraduate Researcher — Computational Astrophysics	May 2025 – June 2026
University of Texas at Dallas – Richardson, TX	

- Developed and optimized Python-based simulations for gravitational lensing, executing large-scale parameter sweeps across high-dimensional spaces to improve computational efficiency
- Applied chi-squared minimization and Bayesian inference to evaluate hundreds of thousands of model configurations, calibrating parameters and identifying sensitivities to input variations
- Implemented numerical ray-tracing algorithms for modeling optical systems under varying condition, extending models toward wave-based effects and using image-based visualization to interpret system behavior
- Presented data-driven results at UTD SPUR 2025; selected for research award
- Tools Used: *Python (NumPy, SciPy, Astropy, Matplotlib, Plotly), MATLAB, Linux, Bash, Jupyter, Git, Glafic*

Restaurant Manager — Family-Owned Business	June 2016 – Present
Golden Deli – Addison, TX	

- Managed operations including transactions, inventory control, and equipment maintenance while troubleshooting issues to maintain efficiency and ensure exceptional customer service

Projects and Builds

Encrypted Password Manager – Secure Storage of Sensitive Data

- Developed a CLI password manager backed by PostgreSQL with encrypted credential storage, account management, configurable security settings, and relational data integrity through foreign-key constraints and cascading deletion
- Implemented AES-256 encryption with Argon2-based key derivation and secure master-password rotation, using atomic database transactions to maintain data integrity and safely re-encrypt stored credentials
- Tools Used: *Python, PostgreSQL, Docker, Git*

Laser Diode Spectroscopy — Hyperfine Transitions of Rubidium

- Diagnosed and corrected a systematic measurement error in a laser setup caused by unintended optical paths in the Michelson interferometer, reducing error from an order-of-magnitude deviation to $< 5\%$
- Developed a fringe-based frequency calibration to extract GHz-scale spectral features and determine hyperfine peak separations consistent with known atomic transitions

Semiconductor Characterization — Hall Effect Measurements

- Determined carrier density, mobility, and conductivity of doped semiconductors by combining Hall effect measurements with transport analysis, validating results against expected material behavior
- Calculated semiconductor band gap (~ 0.75 eV) from temperature-dependent conductivity using $\ln(\sigma)$ vs $1/T$ analysis, achieving agreement within 12% of accepted values

Optical Schlieren Imaging System — Density Gradients in Gas Propulsion

- Designed and built an optical Schlieren imaging system to capture refractive index variations in gas flow, enabling visualization of otherwise undetectable flow structures
- Aligned and optimized optical components (mirrors, a collimated light source, and knife-edge cut-off) to optimize system sensitivity, spatial resolution, and image contrast

Skills

Programming: Java, Python, C++/C/C#, Bash, SQL, HTML/CSS,

Software: Linux (Ubuntu/WSL), Git, MATLAB, Jupyter, PostgreSQL, MS Excel, Docker, AWS, Arduino IDE

Fabrication: Repair and Maintenance, Hand and Power Tools, Basic Electrical Wiring and Circuit Assembly, Soldering